THE NORMALIZATION IN A FINITE TYPE REDUCED SPACE OVER A NAGATA BASE IS FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a reduction to the affine and then to the scheme case; the embedding of global sections along an affine etale cover is the step that lets Noetherianity do the work.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Etale locality of the normalization and of finiteness of the normalization map. Lemma lemma-properties-normalization, used at proof lines 10213-10215. Reduces the claim to an affine base Y = Spec(R).
- Affine etale covers of quasi-compact algebraic spaces. Properties of Spaces, Lemma spaces-properties-lemma-quasi-compact-affine-cover, used at proof lines 10219-10222. Produces the affine scheme U with a surjective etale morphism U -> X.
- The scheme case of Nagata normalization finiteness. Morphisms, Lemma morphisms-lemma-nagata-normalization-finite, used at proof lines 10226-10228. Gives finiteness of the integral closure of R in Gamma(U, O_U).
- Noetherian rings make submodules of finite modules finite. used at proof lines 10223-10226. Transfers finiteness from Gamma(U, O_U) to the subring Gamma(X, O_X).

Source reference. Original lemma 0ABP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13027–13027. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03GR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4538–4538. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03H6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4555–4555. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume that*

(1) *Y is Nagata,*
(2) *f is quasi-separated of finite type,*
(3) *X is reduced.*

Then the normalization $\nu : Y' \to Y$ of Y in X is finite.

Proof. The question is étale local on Y, see Lemma 0ABP. Thus we may assume $Y = \mathrm{Spec}(R)$ is affine. Then R is a Noetherian Nagata ring and we have to show that the integral closure of R in $\Gamma(X, \mathcal{O}_X)$ is finite over R. Since f is quasi-compact we see that X is quasi-compact. Choose an affine scheme U and a surjective étale morphism $U \to X$ (Properties of Spaces, Lemma 03H6). Then $\Gamma(X, \mathcal{O}_X) \subset \Gamma(U, \mathcal{O}_X)$. Since R is Noetherian it suffices to show that the integral closure of R in $\Gamma(U, \mathcal{O}_U)$ is finite over R. As $U \to Y$ is of finite type this follows from Morphisms, Lemma 03GR. □